Homework 2

Company Fundamental Deep Dive

Delivery	Jupyter Notebook (.ipynb)
Dataset	Symbol_Info_extended.csv
Deadline	June 4th

Your Geographic Group Assignment

Find your assigned group using the last digit of your Student ID. You will select your company freely from within that group.

Student ID ends with…	Group	Countries
0 or 1	Group 1 — LATAM & Offshore	Brazil, Mexico, Cayman Islands, Bermuda
2 or 3	Group 2 — Canada	Canada
4 or 5	Group 3 — United Kingdom	United Kingdom
6 or 7	Group 4 — Asia	China, Taiwan, Singapore, South Korea, Hong Kong
8 or 9	Group 5 — Europe	Ireland, Italy, Germany, Netherlands, Luxembourg, Spain, France

Objective

Evaluate the financial condition of one company selected from your assigned geographic group. The analysis must combine financial metrics, peer comparisons, charts, and written interpretation across four business questions: profitability, valuation, growth, and financial strength.

How to Document Your Work

You do not need a markdown cell after every chart. What is required is that your code is commented throughout: explain what each block does, and whenever you make a choice (which metric to use, which industries to keep, which quadrant to focus on) write a short comment explaining why. The only mandatory written section is the Final Summary at the end.

Step 0 — Select a Company

Select one company from your assigned geographic group (see table above). Write a short comment justifying the choice with specific comments on industry position, market cap, profitability, beta, or other aspects.

Step 1 — Data Preparation

Load the dataset and create a working DataFrame containing only the variables required for this homework.

Required identification and peer-selection variables:

- symbol, company_name, sector, industry, country, market_cap

Required financial variables:

- return_on_assets, return_on_equity, profit_margins
- pe_trailing, price_to_book, revenue_growth, debt_to_equity, free_cashflow

Before computing ratios or producing charts, apply the following cleaning rules:

- **Replace infinite values with NaN:** use df.replace([np.inf, -np.inf], np.nan). Infinite values can appear when a ratio is computed from a zero denominator (e.g. dividing by zero revenue or zero equity). They are not valid observations and must not be treated as large numbers.
- **Peer selection by market_cap:** exclude companies with market_cap ≤ 0 before sorting and selecting peers. A non-positive market cap indicates a data quality issue, not a very small company.
- **pe_trailing and price_to_book:** set values ≤ 0 to NaN before any chart or comparison. A negative P/E (loss-making company) and a negative P/B (negative book equity) are not comparable to positive multiples on the same scale and must not appear in valuation charts.

Handling missing values: remove missing values only for the metric used in each specific analysis. If the selected company has no valid value for a metric, state this clearly, plot the available peers, and do not impute the missing value.

Step 2 — Build Peer Groups

For the selected company, build three comparison groups. In every chart, the selected company must be clearly highlighted when its metric value is available. If company_name is missing for any company, use symbol as the label.

Group A — Same Industry

- Select all companies in the same industry as the selected company.
- Sort by market_cap descending and keep the top 10.
- If the selected company is not in the top 10, add it (the group will contain 11 companies).
- If fewer than 10 companies are available in the industry, use all of them.

Group B — Same Geographic Group

- Select all companies that belong to the same geographic group as the selected company (see assignment table).
- Sort by market_cap descending and keep the top 10.
- If the selected company is not in the top 10, add it.

- If fewer than 10 companies are available in the group, use all of them.

Group C — Closest Market Capitalization

- Select the 10 companies with the closest market_cap to the selected company, using absolute distance.
- Sort by market_cap distance and keep the closest 10.
- If the selected company appears in this group, keep only 10 companies total.
- If fewer than 10 valid companies are available, use all of them.

Step 3 — Business Question 1: Is the Company Profitable?

Evaluate profitability using at least two of the following metrics: profit_margins, return_on_assets, return_on_equity.

For each selected metric:

- Clean missing and invalid values for that metric only.
- Compare the selected company with each peer group from Step 2.
- Create one chart per peer group and highlight the selected company when it has a valid value.

Note on return_on_equity: ~26% of companies have negative ROE (due to net losses or negative book equity). This is a valid observation. Ensure your chart accommodates negative values and annotate the zero line.

Required output: at least two profitability metrics, one chart per peer group per metric, and one written interpretation covering: whether the company is above or below the peer median, and whether the metrics are consistent with each other.

Step 4 — Business Question 2: Is the Company Expensive or Cheap?

Evaluate (using only peers from Group A) valuation using pe_trailing and price_to_book (treat values ≤ 0 as missing). Sort charts in ascending order and highlight the selected company.

Required output: one chart for pe_trailing, one for price_to_book, and one written interpretation covering: whether the company appears expensive, cheap, or fairly valued; and whether a low valuation reflects weak fundamentals rather than an opportunity.

Step 5 — Business Question 3: Is the Company Growing?

Evaluate growth using revenue_growth. Compare the selected company with the peer groups from Step 2.

Dataset note — revenue_growth: values are in decimal form (0.08 = +8%). The distribution has heavy tails with outliers above 10 (>1000% growth). If outliers distort the chart scale, apply a cap and state it in a comment.

Required output: one chart per peer group and one written comment covering: whether the company is growing faster or slower than peers, whether growth is positive or negative, and whether it helps explain the valuation in Step 4.

Step 6 — Business Question 4: Is the Company Financially Solid?

Evaluate financial strength using debt_to_equity and free_cashflow (only peers from Group A).

debt_to_equity is expressed as a percentage (e.g. 64 = 64% = 0.64x ratio). Lower values are preferable. The dataset contains outliers above 10,000; if present in your peer group, cap the axis and note it.

free_cashflow is in absolute monetary terms and can be strongly negative. Higher values are preferable. Negative FCF is not automatically a red flag: it may reflect heavy investment in a growth phase.

Required output: one chart for debt_to_equity, one for free_cashflow, and one written interpretation covering: whether the company relies heavily on debt, whether it generates positive free cash flow, and whether the financial profile is strong, weak, or mixed.

Interpretative Reference Ranges

These ranges are indicative only. Financial ratios vary significantly across sectors. A P/E of 12x is normal for a bank, high for a utility, and low for a software company. Always interpret a metric relative to the industry peers in Group A, not against these thresholds in isolation.

Metric	Weak signal	Intermediate	Strong signal	Sector note
profit_margins	< 0%	0 – 10%	> 10%	Retail and transport often < 5%; software and pharma often > 20%
return_on_assets	< 0%	0 – 5%	> 5%	Asset-heavy sectors (utilities, real estate) are structurally lower
return_on_equity	< 0%	0 – 15%	> 15%	Negative ROE can indicate losses or negative book equity — not the same thing
pe_trailing	n/a or < 0	10 – 25x	> 25x (growth premium)	< 10x may signal cheap valuation or weak earnings outlook; sector context is essential

price_to_book	< 1x (distressed?)	1 – 3x	> 3x	Financials and industrials often < 2x; tech and consumer brands often > 5x
revenue_growth	< 0 (declining)	0 – 0.10	> 0.10	Mature sectors (utilities, consumer defensive) typically grow < 5% per year
debt_to_equity*	> 200%	50 – 200%	< 50%	Banks and real estate are structurally more leveraged; compare within industry
free_cashflow	Negative	—	Positive	Negative FCF during a growth phase is not automatically a red flag

* debt_to_equity in this dataset is expressed as a percentage (e.g. 120 = 120% = 1.2x ratio).

Final Summary

Write a final summary in a markdown cell at the end of the notebook covering:

- Why you selected the company and what Homework 1 evidence motivated the choice.
- How the company compares with peers in profitability.
- Whether the company appears expensive, cheap, or fairly valued, and whether growth supports this.
- Whether debt and free cash flow raise concerns.
- Which metrics were unavailable, if any, and how this affected the analysis.